3.2.2 Sulphur dioxide

Preparation

From sulphur: A large-scale production of sulphur dioxide is done by burning sulphur in air. About 6-8% of sulphur is oxidised to SO_3.

$$S + O_2 \longrightarrow SO_2$$
$$2S + 3O_2 \longrightarrow 2SO_3$$

From sulphides: When sulphide ores such as galena (PbS), zinc blende (ZnS) are roasted in air, sulphur dioxide is liberated. Large amounts of sulphur dioxide required for manufacturing of sulphuric acid and other industrial purpose is prepared by this method.

$$2ZnS + 3O_2 \xrightarrow{\Delta} 2ZnO + 2SO_2$$
$$4FeS_2 + 11O_2 \xrightarrow{\Delta} 2Fe_2O_3 + 8SO_2$$

Laboratory preparation: Sulphur dioxide is prepared in the laboratory treating a metal or metal sulphite with sulphuric acid

$$Cu + 2H_2SO_4 \longrightarrow CuSO_4 + SO_2 + 2H_2O$$

$$SO_3^{2-} + 2H^+ \longrightarrow H_2O + SO_2$$

Properties:

Sulphur dioxide gas is found in volcanic eruptions. A large amount of sulphur dioxide gas is released into atmosphere from power plants using coal and oil and copper melting plants. It is a colourless gas with a suffocating odour. It is highly soluble in water and it is 2.2 times heavier than air. Sulphur dioxide can be liquefied (boiling point 263 K) at 2.5 atmospheric pressure and 288 K.

Chemical properties

Sulphur dioxide is an acidic oxide. It dissolves in water to give sulphurous acid.

$$SO_2 + H_2O \rightleftharpoons \underset{\text{Sulphurous acid}}{H_2SO_3}$$
$$H_2SO_3 \rightleftharpoons 2H^+ + SO_3^{2-}$$

Reaction with sodium hydroxide and sodium carbonate: Sulphur dioxide reacts with sodium hydroxide and sodium carbonate to form sodium bisulphite and sodium sulphite respectively.

$$SO_2 + NaOH \longrightarrow \underset{\text{Sodium bisulphite}}{NaHSO_3}$$
$$2SO_2 + Na_2CO_3 + H_2O \longrightarrow 2NaHSO_3 + CO_2$$
$$2\,NaHSO_3 \longrightarrow \underset{\text{Sodium sulphite}}{Na_2SO_3} + H_2O + SO_2$$

Oxidising property: Sulphur dioxide, oxidises hydrogen sulphide to sulphur and magnesium to magnesium oxide.

$$2H_2S + SO_2 \longrightarrow 3S + 2H_2O$$
$$2Mg + SO_2 \longrightarrow 2MgO + S$$

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